CLOCK SIGNALS ▪ RAPIDITY ▪ MEDIANTS

Doppler shift to the twin paradox

Two reciprocal Doppler factors; one sum of elapsed times.

1 Start with a receding siren

A siren moves away at speed v through still air. The listener rests in the air. Let u be the sound speed and $P_0 = 1/f_0$ the emitted period.

$$f_{\text{rec}} = f_0 \frac{u}{u+v}, \qquad \frac{P_{\text{rec}}}{P_0} = 1 + \frac{v}{u}.$$

During each emitted period, the siren moves vP_0. That adds flight time vP_0/u:

$$P_{\text{rec}} = P_0 + \frac{vP_0}{u}.$$

For approach, replace $u+v$ by $u-v$, with $v < u$.

2 Light and Bondi's k

For light, periods are measured on the emitter's and receiver's own clocks. Write $\beta = v/c = \tanh\eta$, where η is rapidity. For recession, define

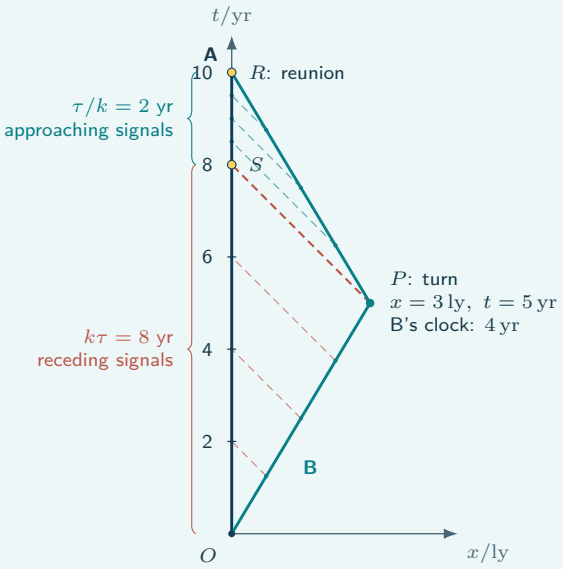
$$k = \frac{P_{\text{rec}}}{P_0} = \sqrt{\frac{1+\beta}{1-\beta}} = e^{\eta} > 1.$$

Motion	P_{rec}/P_0	f_{rec}/f_0
Receding	k	k^{-1}
Approaching	k^{-1}	k

Reversing rapidity exchanges k and $1/k$.

3 One diagram: read the traveller's clock by light

A stays at $x = 0$; B coasts out and back, taking proper time τ per leg. Both send ticks at proper frequency f_0. Work in flat spacetime with an idealised instantaneous turn.



Dashed $45°$ lines carry B's yearly ticks to A. The turnaround message reaches A at S, year 8.

Add the two receiving intervals.

B's leg	B's elapsed time	A receives over
Outbound	τ	$k\tau$
Return	τ	τ/k

$$T_A = k\tau + \frac{\tau}{k}, \qquad T_B = 2\tau.$$

Thus the ratio of their elapsed ages is

$$\boxed{\frac{T_A}{T_B} = \frac{k+k^{-1}}{2} = \cosh\eta = \frac{1}{\sqrt{1-\beta^2}} = \gamma}.$$

The age difference is

$$T_A - T_B = \tau\,\frac{(k-1)^2}{k} \geq 0.$$

The example in the diagram

$$\beta = \frac{3}{5}, \qquad k = 2, \qquad \tau = 4\,\text{yr}.$$

$$T_A = 2(4) + \frac{4}{2} = 10\,\text{yr}, \qquad T_B = 2(4) = 8\,\text{yr}.$$

Earth ages 10 years; the traveller ages 8.

4 Why their views differ

A stays in one inertial frame; B changes to the inbound frame at P. B's received Doppler rate switches at P; A receives the news at S.

	A receives B	B receives A
Receding rate/f_0	$1/k$	$1/k$
For duration	$k\tau$	τ
Approaching rate/f_0	k	k
For duration	τ/k	τ

Durations are measured on the receiver's own clock.

For one tick per year in the example: B receives $2+8=10$ Earth ticks; A receives $4+4=8$ traveller ticks. At reunion both have the complete clock record.

5 The mediant of tick records

Combine actual tick counts N_i and receiving durations D_i by their mediant:

$$\frac{N_1}{D_1}, \frac{N_2}{D_2} \quad \longmapsto \quad \frac{N_1+N_2}{D_1+D_2}.$$

Earth's average received rate is therefore

$$\frac{\overline{f}_{B\to A}}{f_0} = \frac{\tau+\tau}{k\tau+\tau/k} = \frac{2}{k+k^{-1}} = \gamma^{-1}.$$

For the traveller,

$$\frac{\overline{f}_{A\to B}}{f_0} = \frac{\tau/k+k\tau}{2\tau} = \gamma.$$

A: harmonic mean, from equal tick counts. **B: arithmetic mean**, from equal receiving durations.

Reading: OpenStax: sound Doppler; light Doppler; M. Moriconi: Doppler approach, Sections 2 and 6; Einstein Online: travelling twins.